Reg. No. | | | | | | | | | | |

[16CSS04605]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE–2019

ADVANCED ANDROID PROGRAMMING

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any five full questions selecting atleast two from each part.

PART - A

1. a. List and explain the different types of sensors available in different smartphones. **06**

   b. Discuss the need of setting permissions in manifest file and list and explain the permissions required for the following features. **08**

   i. Bluetooth ii. Wi-Fi iii. GPS iv. SMS

2. a. What is the use of Broadcast Receivers in Android? Explain with suitable code. **08**

   b. Implement an application that creates an alert upon receiving a message. **06**

3. a. Implement the following using Messenger API for Server: **08**

   i. A command to display "Hello" Message

   ii. Handler for handling incoming commands/messages

   iii. Return an IBinder object

   b. What are Sockets in android? How do you establish socket connection between two devices/applications in the same mobile? **06**

4. a. Compare Process and Threads. With a neat Sketch, detail the different states of a thread. **08**

   b. Demonstrate through a program how the outputs/results are published on to UI thread using AsyncTask Class. **06**

PART - B

5. a. Write an application that draws basic graphical primitives (rectangle,circle) on the screen. **05**

   b. With suitable code, explain the role of AnimationDrawable Class and to create Frame Animation in Android. **09**

6. a. Discuss the role of JSON and name its classes in Android. Explain JSON Parsing with suitable code. **08**

   b. What are Webservices? Discuss Android REST Web Service. **06**

7. a. Define Fragments. With a neat diagram, explain life cycle of Fragment in Android. **09**

   b. Discuss Android Fragment Transition in Detail. **05**

8. a. Discuss Android Themes with examples. **06**

   b. Explain the need of Card-Based Layout in Android. How do you create Cards? **08**

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